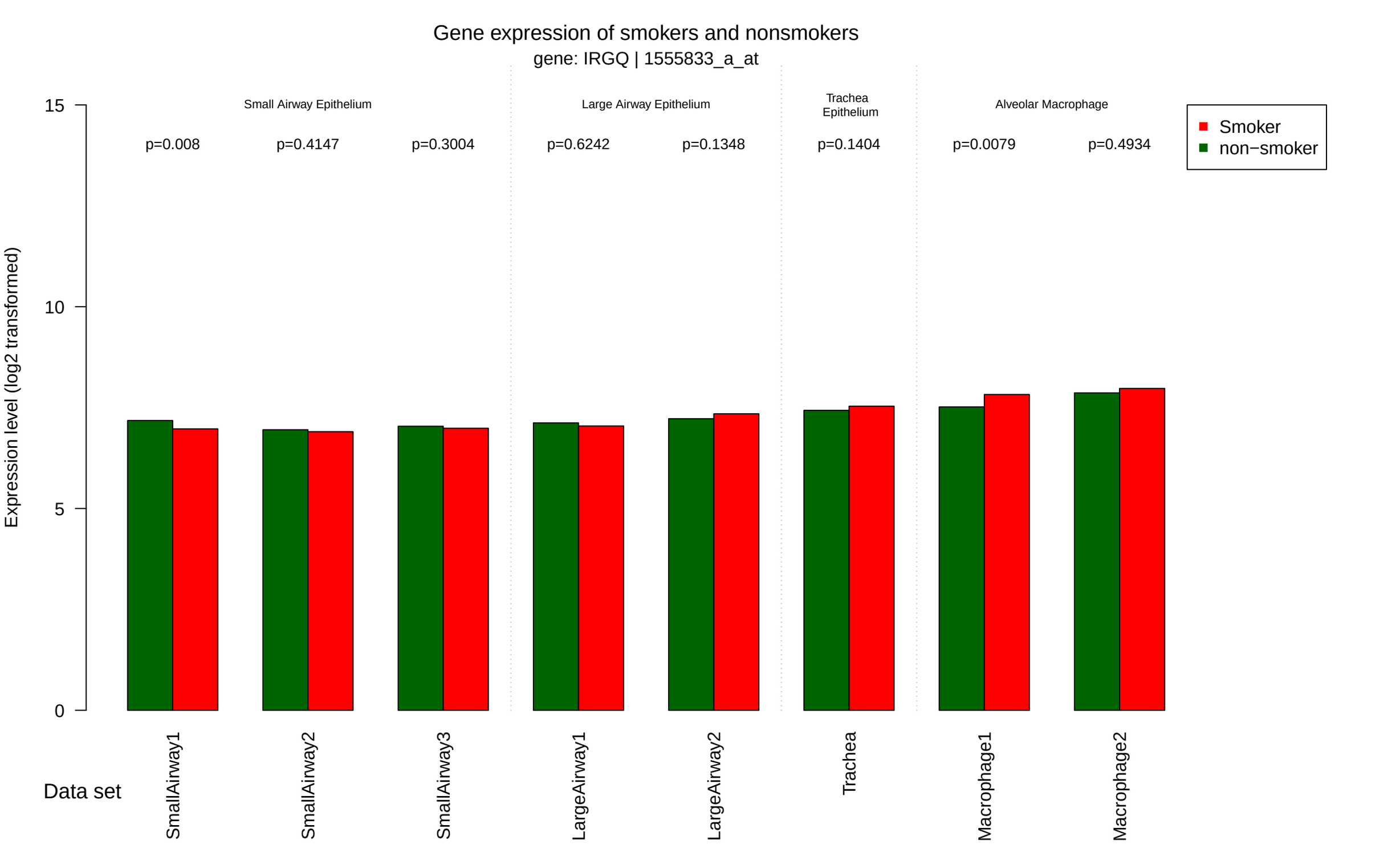
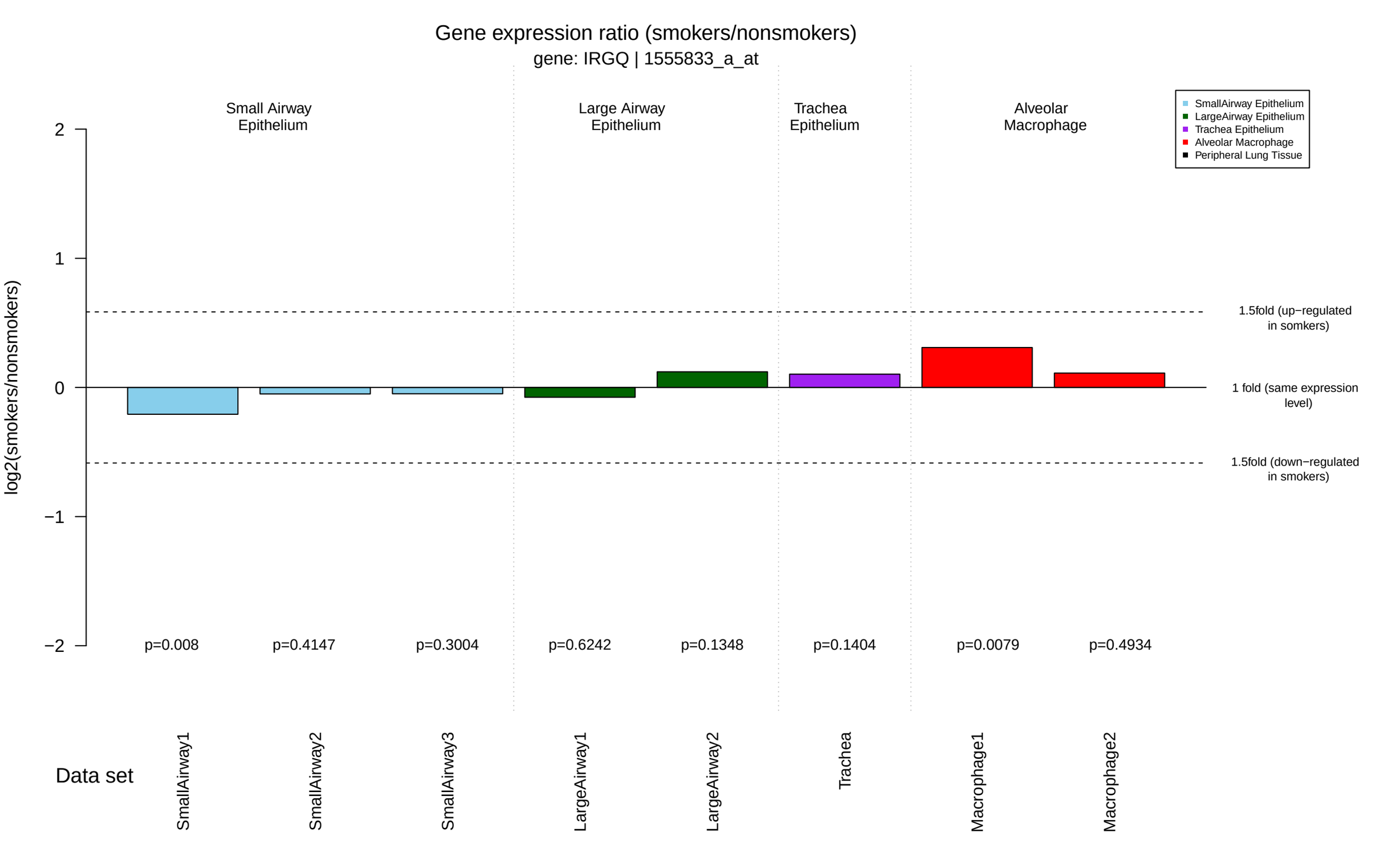
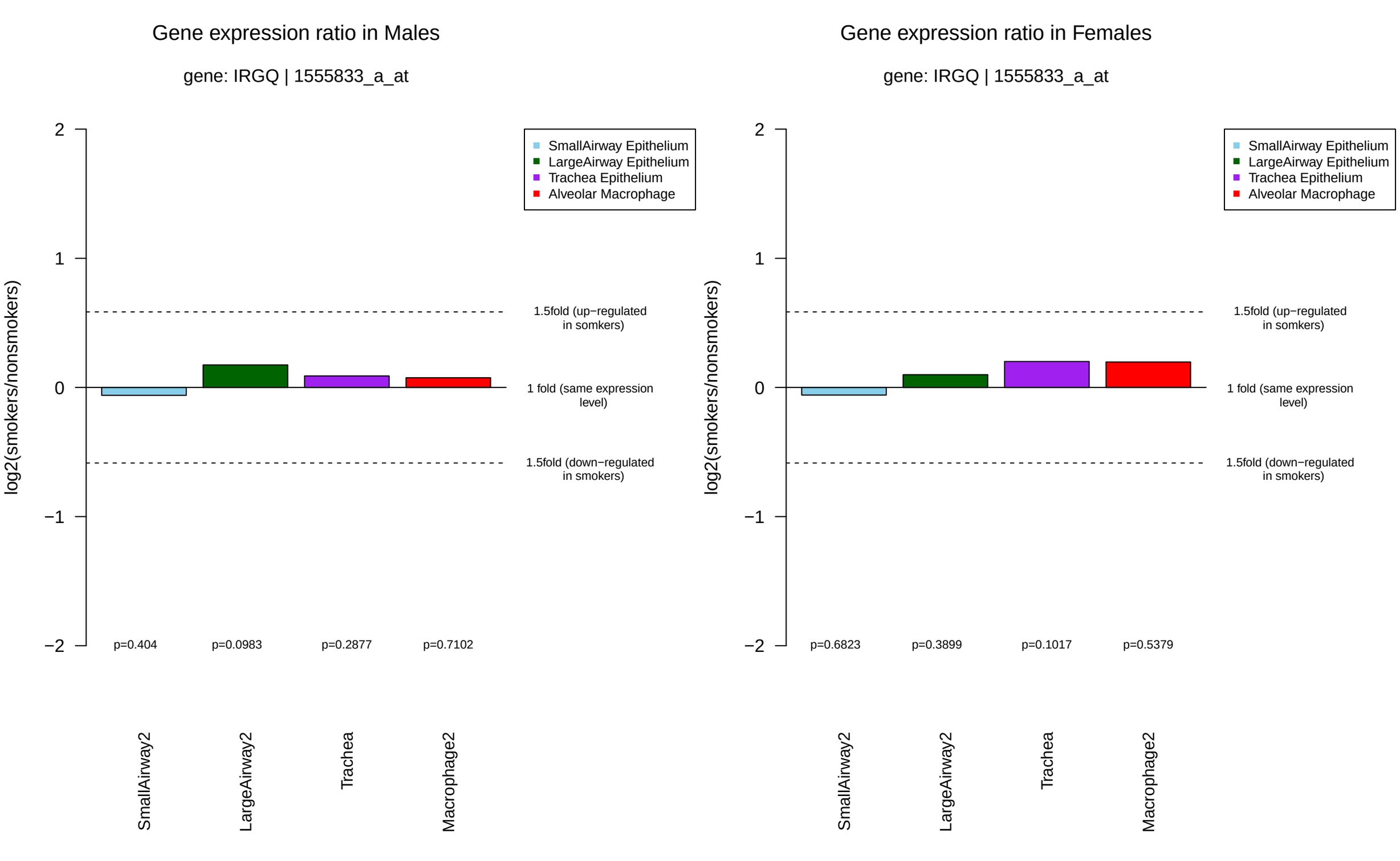
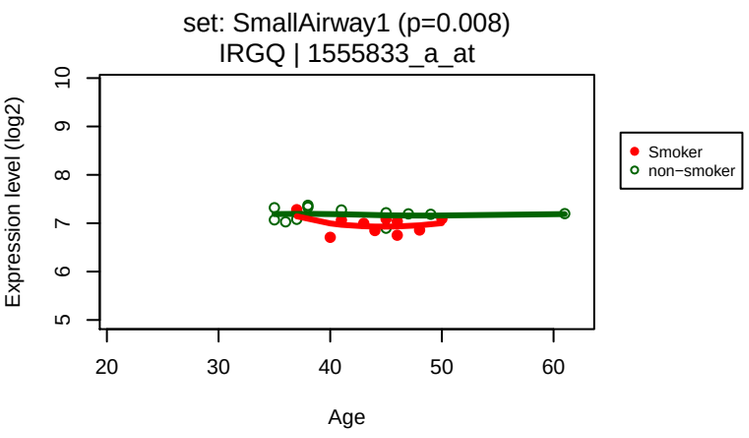
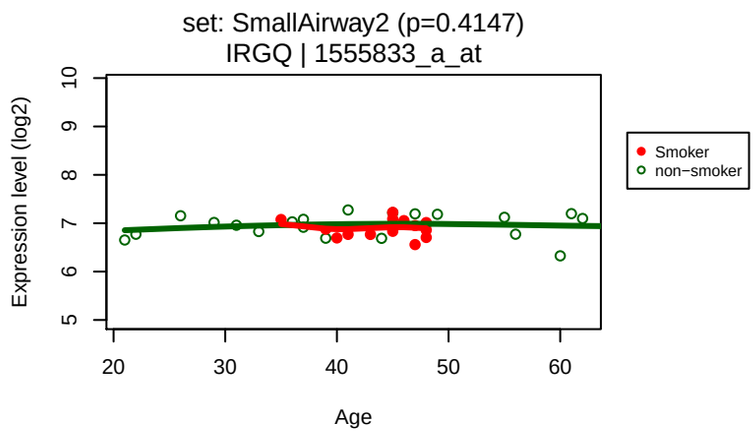
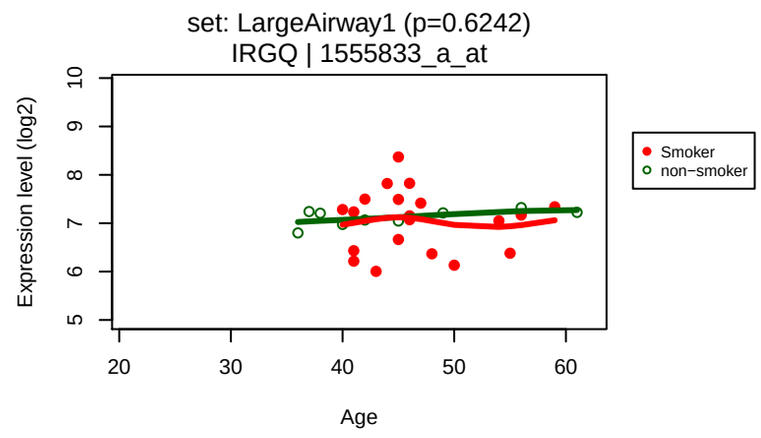

Gene expression of smokers and nonsmokers
gene: IRGQ | 1555833_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.008
p=0.4147
p=0.3004
p=0.6242
p=0.1348
p=0.1404
p=0.0079
p=0.4934
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

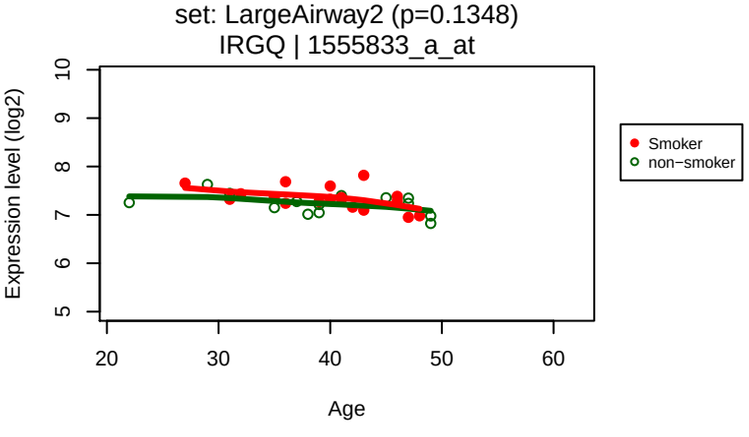

Gene expression ratio (smokers/nonsmokers)
gene: IRGQ | 1555833_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.008
p=0.4147
p=0.3004
p=0.6242
p=0.1348
p=0.1404
p=0.0079
p=0.4934
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

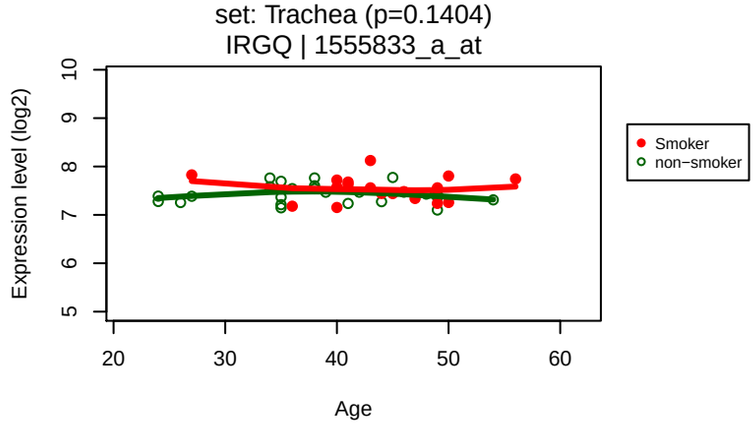

Gene expression ratio in Males
gene: IRGQ | 1555833_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.404
p=0.0983
p=0.2877
p=0.7102
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRGQ | 1555833_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6823
p=0.3899
p=0.1017
p=0.5379
SmallAirway2
LargeAirway2
Trachea
Macrophage2

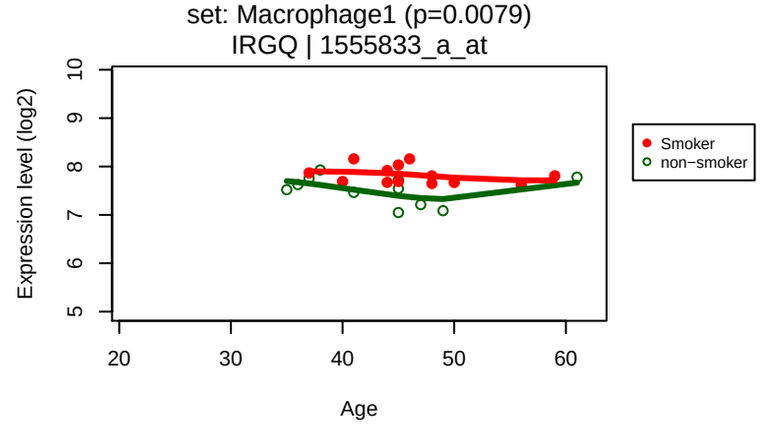

set: SmallAirway1 (p=0.008)
IRGQ | 1555833_a_at
Expression level (log2)
Age
Smoker
non−smoker

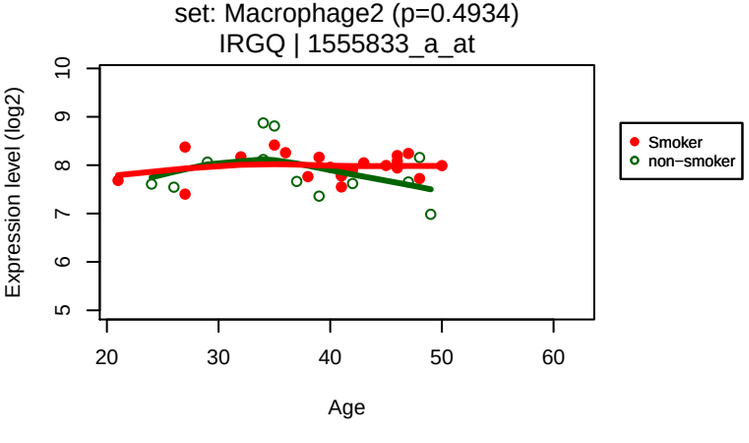

set: SmallAirway2 (p=0.4147)
IRGQ | 1555833_a_at
Expression level (log2)
Age
Smoker
non−smoker

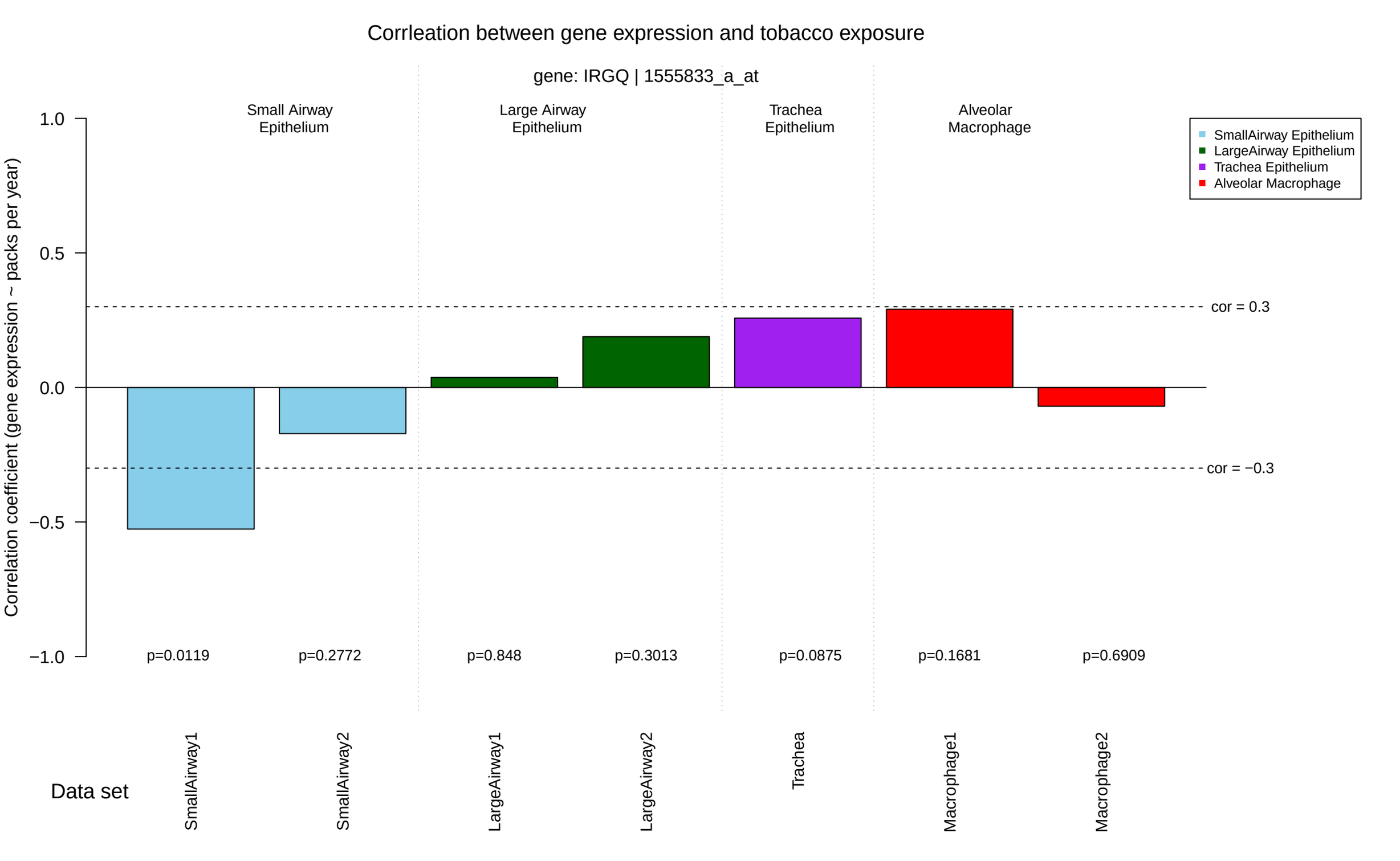

set: LargeAirway1 (p=0.6242)
IRGQ | 1555833_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1348)
IRGQ | 1555833_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1404)
IRGQ | 1555833_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0079)
IRGQ | 1555833_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4934)
IRGQ | 1555833_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRGQ | 1555833_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0119
p=0.2772
p=0.848
p=0.3013
p=0.0875
p=0.1681
p=0.6909
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2